| Field | Entry |
|---|---|
| Source ID | WBC-0029 |
| Title | In Situ Construction of Thermotropic Shape Memory Polymer in Wood for Enhancing Its Dimensional Stability |
| Source type | Paper |
| Link / File | https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8876273/ |
| Craft applicability | Wood (moisture-prone species — Noni and Albizia lebbeck used in Kallakurichi carving) |
| Primary Why-theme | 10. Failure, Repair and Quality |
| Process stage | Material augmentation (pre/post-carving treatment) |
| Questions answered | Why does vacuum-pressure resin impregnation reduce warping and cracking in moisture-prone wood species? |
| Evidence level | E3 |
| Relevance note | Demonstrates that vacuum-pressure impregnation of resin into wood cell lumens significantly enhances dimensional stability and reduces cracking, supporting the report's proposal to use vacuum-aided resin infusion on moisture-prone species like Noni and Albizia lebbeck. |